

Bayesian Forecasting of Household Electricity Demand

Project 17 — methods, modelling decisions, and obstacles

Technical working document

September 15, 2026

Abstract

This document records what the project asks for, how the data were prepared, what model was built and why, which algorithms were implemented to fit it, and — at equal length — what went wrong along the way. Four years of one-minute electricity measurements from a single French household are aggregated to 1,417 usable daily consumption totals and modelled with a Bayesian harmonic regression carrying a linear trend, annual harmonics at period 365 days, day-of-week effects, and a heavy-tailed autoregressive error process. Inference uses a purpose-built Gibbs sampler with Metropolis steps, validated against simulated data with known truth. Seven specifications are compared by WAIC and, more importantly, by genuinely held-out forecasts over a twelve-month test period. Several conclusions reverse the choices that seemed obvious at the outset: the log transform hurts rather than helps, an AR(2) error process is favoured by WAIC but rejected out of sample, and a plain day-of-year climatology still outperforms every fitted model at long horizons. The last of these remains open and is documented as such.

1 What the project asks

1.1 The brief

The assignment supplies a dataset, mandates one component of the model, and poses six questions. In its own terms: characterise temporal patterns in daily household electricity consumption, and develop a Bayesian analysis usable for predicting future daily demand.

The seasonal component is not a free choice. The model *must* represent annual seasonality with sine and cosine terms at period 365 days, beginning with the first harmonic, $\sin(2\pi t/365)$ and $\cos(2\pi t/365)$, and the analysis must investigate whether adding the second harmonic, $\sin(4\pi t/365)$ and $\cos(4\pi t/365)$, improves the representation.

1.2 The six questions, and what each one forces into the model

The questions read like prose but function as a specification. Each names a term the model must contain and a posterior that must be reported.

Question	Required model term	Required output
Q1. How does consumption vary over the study period?	Nothing new; descriptive	Series plot, summary statistics, variance decomposition by component
Q2. Is there a long-term increase or decrease?	Linear trend in t	Posterior of the slope in kWh/year and %/year, with $\Pr(\text{slope} < 0 \mid \text{data})$
Q3. What annual pattern is present, and does harmonic 2 improve on harmonic 1?	Harmonics (mandated)	Amplitude and phase in kWh; a defended comparison of $J=1$ against $J=2$
Q4. Are there systematic weekly patterns?	Day-of-week effects	Seven posterior deviations with intervals; is the weekend genuinely different?

Question	Required model term	Required output
Q5. How accurately can demand be predicted, and how does uncertainty change with horizon?	A serially correlated error process, plus honest validation	Predictive interval width as a function of h ; held-out accuracy and coverage
Q6. Are there periods not adequately explained by the model?	Nothing new, but requires posterior predictive checking	A dated table of anomalies with interpretation

Two consequences follow immediately. First, trend, harmonics, day-of-week effects, and a correlated error term are all mandatory — omit any one and a question becomes unanswerable. Second, Q5 is the load-bearing question, and it is the one that rules out the simplest error assumption: if errors were independent, predictive uncertainty would be identical one day ahead and three hundred days ahead, which is both false and exactly what the question probes.

1.3 Why this is a Bayesian problem

The word “uncertainty” appears three times in the brief. A frequentist could fit this regression in one line. What the assignment wants is a posterior distribution for every quantity and a full predictive distribution for every future day, so that statements like “the annual cycle swings 16.2 kWh/day, 95% CI [14.6, 17.7]” and “there is a 77.5% posterior probability that consumption is declining” are available directly rather than through an appeal to asymptotics.

2 The data

2.1 Provenance and physical form

The UCI *Individual Household Electric Power Consumption* dataset records 2,075,259 one-minute measurements from a house in Sceaux, about 7 km from Paris, between December 2006 and November 2010 (47 months). It was donated by Georges Hébrail and Alice Bérard of EDF R&D, the French national utility. This is *one house*, not a grid — a point that governs how every conclusion must be worded.

The file `household_power_consumption.txt` is semicolon-delimited, roughly 127 MB uncompressed, with nine columns:

#	Column	Unit	Content
1	Date	dd/mm/yyyy	European format, day first
2	Time	hh:mm:ss	Local time
3	Global_active_power	kW	The raw material: mean real power that minute
4	Global_reactive_power	kW	Reactive power; unused
5	Voltage	V	Near 240; unused
6	Global_intensity	A	Nearly collinear with column 3; unused
7	Sub_metering_1	Wh	Kitchen (dishwasher, oven, microwave)
8	Sub_metering_2	Wh	Laundry (washer, drier, fridge, light)
9	Sub_metering_3	Wh	Electric water heater and air conditioner

The three sub-meters deliberately do not sum to the total; the remainder (lighting, electronics, heating) is the bulk of consumption and is used here only as a validation check.

2.2 Four traps that silently corrupt the pipeline

1. **Missing values are the literal character ?**, not blanks. Without declaring it, every numeric column parses as text and daily totals come out as garbage or raise errors.
2. **Dates are day-first**. Parsing as `%m/%d/%Y` mangles the first twelve days of every month and fails on the rest.
3. **Missingness is contiguous**. About 1.252% of minutes (25,979 of 2,075,259) lack a power reading, and they arrive in runs, not scattered singletons, so individual days can lose hundreds of consecutive minutes.
4. **The boundary days are partial**. The record starts mid-afternoon on 16 Dec 2006 and ends in the evening on 26 Nov 2010. Both were dropped.

2.3 Aggregation: a sum divided by sixty

`Global_active_power` is *power* in kilowatts averaged over one minute. Energy is power multiplied by time, and one minute is $1/60$ of an hour, so the daily total is

$$y_d = \frac{1}{60} \sum_{m \in \text{day } d} \text{GAP}_m \quad [\text{kWh}]. \quad (1)$$

This is a sum divided by 60, not a mean. Treating it as a mean understates daily consumption by a factor of $1440/60 = 24$, and it is the single most common error made with this dataset.

2.4 The missingness rule

Naively applying (1) to a day missing 400 minutes manufactures an artificially low day. Because Q6 asks which days are anomalous, such artefacts would be reported as findings. The rule adopted is therefore:

- compute $n_{\text{obs}}(d)$, the count of non-missing minutes out of 1440;
- retain the day only if $n_{\text{obs}}(d) \geq 0.95 \times 1440$;
- otherwise record the outcome as missing — do not rescale, do not zero-fill;
- keep the day on the calendar grid regardless, so the error process is defined on every date.

A validation check confirms parsing: sub-metered energy must never exceed the whole-house total. Violations: 0 of 1,417 days.

2.5 The resulting series

Calendar days on grid	1,440	Mean daily consumption	26.19 kWh
Days retained (coverage $\geq 95\%$)	1,417	Standard deviation	10.00 kWh
Days recorded missing	23	Minimum	4.17 kWh (25 Aug 2008)
of which zero observed minutes	9	Maximum	79.56 kWh (23 Dec 2006)
Sub-meter validation failures	0	Median	25.94 kWh

Annualised consumption is stable and plausible for a French home with an electric water heater: 9,796 (2007), 9,389 (2008), 9,452 (2009), and 9,284 kWh/year (2010, partial). Monthly means range from 13.67 kWh in August to 35.64 kWh in December.

3 Exploratory analysis that determined the model

Rather than assert a specification, three pre-fits were used to choose one. All use ordinary least squares on the same mean structure (intercept, linear trend, two harmonics, day-of-week) and

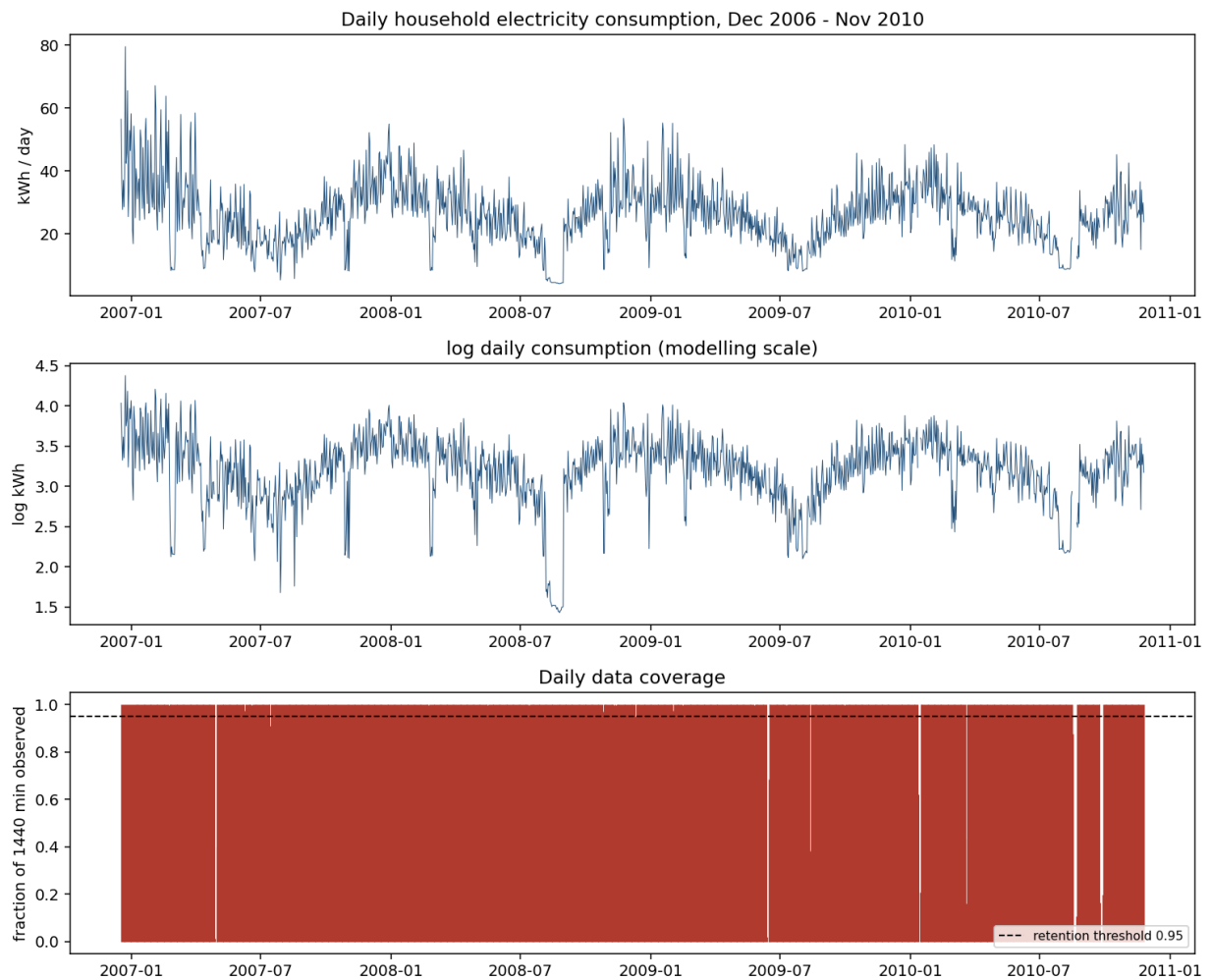


Figure 1: The daily series on both scales, and daily data coverage against the 95% retention threshold. The contiguous nature of the gaps is visible in the bottom panel; the deep summer troughs in the top panel are real absences, not missing data.

examine the residuals.

3.1 Response scale: the log transform is the wrong default

Scale	R^2	Residual skew	Residual kurtosis	Heterosced. p	$\hat{\rho}_1$
y (kWh)	0.452	0.059	5.00	< 0.0001	0.368
$\sqrt{y}$	0.445	-0.699	4.57	0.0239	0.469
$\log y$	0.410	-1.450	6.18	< 0.0001	0.583

The conventional argument for logging a positive, right-skewed response fails here. Raw daily kWh, once season and day-of-week are removed, is *already* symmetric (skew 0.059). Taking logs overcorrects badly (skew -1.450), because the multi-week absence periods — August 2008 sits at 4.76 kWh/day for 25 consecutive days — become extreme *left* outliers on the log scale. The transform creates the asymmetry it was meant to remove. Gaussian kurtosis is 3; every scale is well above it, which is the argument for a t likelihood regardless of scale.

3.2 Error memory: how much, and is it real?

The residual autocorrelation function decays far more slowly than AR(1) implies. The observed values at lags 1, 2 and 3 are 0.582, 0.455 and 0.432, and lag 7 is still 0.273 — where an AR(1) with $\rho_1 = 0.582$ would predict only 0.023. Taken at face value this argues for AR(3) or higher.

It should not be taken at face value. Repeating the exercise with the four sustained absence blocks removed changes the answer completely:

Sample	BIC-preferred order	$\hat{\phi}_1$	$\hat{\sigma}_\eta$	marginal sd / one-step sd
All observed days	$p = 3$	0.453	0.270	1.25
Absence blocks excluded	$p = 1$	0.380	0.258	1.09

The apparent long memory is largely a *regime* effect: a three-week absence looks like enormous persistence to an autocorrelation function. The right response is not a higher-order AR but AR(1) for genuine weather persistence plus a heavy-tailed innovation to absorb the blocks — a conclusion the out-of-sample comparison later confirms.

The four blocks are: 24 Feb–2 Mar 2007 (7 days, mean 9.18 kWh); 6–30 Aug 2008 (25 days, 4.76 kWh); 1–8 Aug 2009 (8 days, 9.01 kWh); and 29 Jul–14 Aug 2010 (17 days, 9.09 kWh).

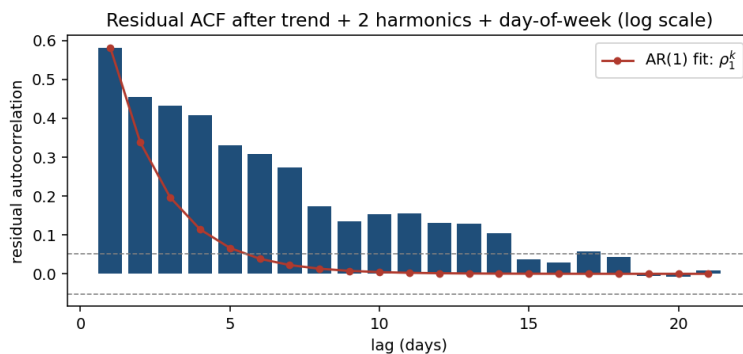


Figure 2: Residual autocorrelation after trend, two harmonics, and day-of-week, against the geometric decay an AR(1) would produce. The gap at lags 3–10 is what the absence blocks contribute.

4 The model

4.1 Specification

Let y_t denote daily consumption in kWh on a gap-free daily grid $t = 1, \dots, T$ with $T = 1440$, and let $z_t = g(y_t)$ for a link g that is either the identity or the logarithm (the choice is treated as a modelling decision to be tested, not assumed). Then

$$z_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \quad (2)$$

$$\mathbf{x}_t^\top \boldsymbol{\beta} = \beta_0 + \beta_1 \frac{t - \bar{t}}{365} + \sum_{j=1}^J \left[a_j \sin \frac{2\pi j t}{365} + b_j \cos \frac{2\pi j t}{365} \right] + \sum_{k=2}^7 \delta_k \mathbf{1}\{\text{dow}_t = k\}, \quad (3)$$

$$\varepsilon_t = \sum_{\ell=1}^p \phi_\ell \varepsilon_{t-\ell} + \eta_t, \quad \eta_t \sim t_\nu(0, \sigma^2). \quad (4)$$

Monday is the baseline day, so δ_k measures the deviation of day k from Monday; the reported weekly profile is re-centred on the weekly mean. The trend is centred at $\bar{t}$, which decorrelates it from the intercept and makes β_0 the level at the midpoint of the record. The harmonics use raw t , as mandated.

Three features of (4) are deliberate and each earns its place in Section 5.6.

The AR term is what makes Q5 answerable. For an AR(1), the h -step predictive standard deviation is $\sigma \sqrt{\sum_{j<h} \phi^{2j}}$, rising from σ at $h = 1$ toward the stationary limit $\sigma/\sqrt{1 - \phi^2}$. The ratio of those two numbers *is* the answer to “how does uncertainty change with the horizon”. With independent errors the ratio is 1 and the question is vacuous.

Heavy tails absorb the absence regime. With Gaussian innovations, the four absence blocks inflate σ and blur every other inference. The t_ν innovation lets the model treat them as outliers rather than as evidence about typical variability.

The error process lives on the full calendar grid. ε_t is defined for all t , but the likelihood is evaluated only on observed days. The 23 missing days are thus *imputed* as part of inference rather than dropped, which both preserves the AR structure across gaps and propagates the uncertainty about those days into everything else.

4.2 Priors

Priors are weakly informative and scaled to the response, which matters because the same code fits both the log scale (coefficients of order 0.1) and the kWh scale (coefficients of order 10).

Parameter	Prior	Rationale
β_0	$N(0, \tau_0^2)$, $\tau_0 = 5$ (log) or 100 (kWh)	Generous relative to a level of 3.2 or 26
other β	$N(0, \tau^2)$, $\tau = 2$ (log) or 40 (kWh)	Admits effects far larger than any plausible one
σ	half- $t(3, 0, A)$, $A = 1$ (log) or 20 (kWh)	Gelman’s recommendation: heavy-tailed, proper, no spike at 0
ϕ	Uniform on the stationary region	Non-informative but excludes non-stationary forecasts
$\nu - 2$	Exponential, mean 10	Keeps the variance finite ($\nu > 2$) while allowing near-Gaussian tails
z_t , t missing	Implied by (4)	Imputation, not deletion

The half- t prior on σ is implemented through the standard inverse-gamma auxiliary variable: if $\sigma^2 \mid \xi \sim \text{IG}(\nu_\sigma/2, \nu_\sigma/(2\xi))$ and $\xi \sim \text{IG}(1/2, 1/A^2)$, then $\sigma \sim \text{half-}t(\nu_\sigma, 0, A)$ and both steps stay conjugate.

5 Algorithms and techniques

5.1 Why a hand-written sampler

No probabilistic programming language was available in the environment (no JAGS, Stan, PyMC, or R), so the sampler was written from scratch in `numpy/scipy`. This turned out to be an advantage rather than a constraint: the model is conditionally conjugate in almost every block, so a Gibbs sampler with two Metropolis steps mixes better than generic HMC would on the 1,440-dimensional latent error path, and it runs a four-chain, 40,000-iteration fit in about 44 seconds.

5.2 The t distribution as a scale mixture

The key device is to write the t_ν innovation as a continuous mixture of normals,

$$\eta_t \mid \lambda_t \sim N(0, \sigma^2/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2), \quad (5)$$

so that marginalising λ_t recovers $\eta_t \sim t_\nu(0, \sigma^2)$. Conditional on $\boldsymbol{\lambda}$ the model is a weighted Gaussian regression, which restores conjugacy for $\boldsymbol{\beta}$ and σ^2 . The λ_t have a second, interpretive use: a small posterior mean λ_t marks a day the likelihood had to downweight.

5.3 Full conditionals

Write $\tilde{\mathbf{x}}_t = \mathbf{x}_t - \sum_\ell \phi_\ell \mathbf{x}_{t-\ell}$ and $\tilde{z}_t = z_t - \sum_\ell \phi_\ell z_{t-\ell}$ for the quasi-differenced data, and let $w_t = \lambda_t/\sigma^2$.

$\boldsymbol{\beta}$ (Gibbs, conjugate normal). Exactly

$$\boldsymbol{\beta} \mid \cdot \sim N(\mathbf{Q}^{-1}\mathbf{r}, \mathbf{Q}^{-1}), \quad \mathbf{Q} = \boldsymbol{\Sigma}_0^{-1} + \sum_{t>p} w_t \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t^\top + \mathbf{X}_{1:p}^\top \mathbf{V}_0^{-1} \mathbf{X}_{1:p},$$

with $\mathbf{r}$ the matching linear term. The third piece is the contribution of the first p days, drawn from the *exact* stationary distribution $\mathbf{V}_0$ of the $\text{AR}(p)$ rather than conditioned away, so the chain starts stationary and no warm-up of the error process is required. For $p = 1$, $\mathbf{V}_0 = \sigma^2/(1 - \phi^2)$; for $p = 2$ it is obtained by solving the discrete Lyapunov equation $\mathbf{V} = \mathbf{A}\mathbf{V}\mathbf{A}^\top + \mathbf{Q}$ in companion form.

σ^2 and ξ (Gibbs, inverse-gamma). $\sigma^2 \mid \cdot \sim \text{IG}((\nu_\sigma + T)/2, \nu_\sigma/(2\xi) + S/2)$ where $S = \sum_{t>p} \lambda_t \eta_t^2 + \boldsymbol{\varepsilon}_{1:p}^\top \mathbf{V}_0^{-1} \boldsymbol{\varepsilon}_{1:p} \sigma^2$, followed by $\xi \mid \sigma^2 \sim \text{IG}((\nu_\sigma + 1)/2, A^{-2} + \nu_\sigma/\sigma^2)$.

λ_t (Gibbs, gamma). $\lambda_t \mid \cdot \sim \text{Gamma}((\nu + 1)/2, (\nu + \eta_t^2/\sigma^2)/2)$, vectorised across all t .

ν (Metropolis, random walk on $\log(\nu - 2)$). The conditional depends on the data only through $\boldsymbol{\lambda}$, giving the gamma likelihood $\prod_t \text{Gamma}(\lambda_t; \nu/2, \nu/2)$; the Jacobian $\log(\nu - 2)$ is included in the acceptance ratio.

ϕ (Metropolis, random walk). The conditional is Gaussian in ϕ multiplied by the stationary-initialisation factor

$$|\mathbf{V}_0|^{-1/2} \exp\left\{-\frac{1}{2} \boldsymbol{\varepsilon}_{1:p}^\top \mathbf{V}_0^{-1} \boldsymbol{\varepsilon}_{1:p}\right\},$$

which is not conjugate. Proposals outside the stationary region are rejected outright by checking that the companion matrix has spectral radius below one.

Missing z_t (Gibbs, normal). The log conditional is quadratic in ε_t , so its exact mean and precision are read off the gradient and curvature of $\sum_s \lambda_s \eta_s^2$ with respect to ε_t , accounting for the $p + 1$ terms in which ε_t appears.

Both random-walk step sizes are adapted during burn-in toward a 30% acceptance rate and then held fixed, so the sampled portion of the chain uses a time-homogeneous kernel.

5.4 Convergence diagnostics

Rank-normalised split- $\hat{R}$ and bulk effective sample size were implemented directly, following Vehtari et al.: draws are rank-normalised through the inverse normal CDF, each chain is split in half to expose within-chain drift, and ESS uses Geyer’s initial positive sequence to truncate the autocorrelation sum. Targets were $\hat{R} < 1.01$ and ESS > 400 for every parameter.

5.5 Validation against simulated data

A sampler that has not been tested against known truth cannot be trusted. Data were simulated from the model itself with known β , ϕ , σ , and ν , and with missing days blanked in contiguous blocks to mimic the real pattern. Result: 14 of 15 parameters had the true value inside the 95% credible interval (the miss was δ_{Tue} , truth 0.060 against an interval ending at 0.059), worst $\hat{R} = 1.014$, minimum bulk ESS 317. Recovery of ϕ was essentially exact: truth 0.450, posterior mean 0.449, interval $[0.406, 0.492]$.

5.6 Model comparison machinery

WAIC. Computed from pointwise log predictive densities using the one-step-ahead conditional factorisation $\sum_t \log p(y_t \mid y_{<t}, \theta)$. Two details make the comparison valid. First, models with different AR orders would otherwise use different numbers of terms, so a *single shared index set* is used throughout: the 1,399 days that are observed and whose two preceding days are also observed. Second, log-scale models carry the Jacobian $-\log y_t$, so every log-likelihood is a density of daily kWh and models on different response scales are directly comparable.

Held-out forecasting, two designs. WAIC scores one-step-ahead fit, which is the wrong target for a forecasting question, so every modelling decision was re-tested out of sample. Models were refitted on data through 30 Nov 2009 (1,072 observed days) and never saw the final twelve months (345 observed days).

- **Design A — single origin.** A twelve-month forecast from 30 Nov 2009. This is the honest “what could you have said a year in advance” test and it produces the fan chart and coverage table. Its weakness is that the 361 horizons come from one realisation, so horizon-by-horizon scores are correlated.
- **Design B — rolling origin.** Horizons 1–60 from origins spread through the holdout year. Parameters still come only from the training period, but each forecast conditions on actuals up to its own origin. Crucially this requires *no refitting*: given β , the error at any day with an observed response is simply $\varepsilon_t = z_t - \mathbf{x}_t^\top \beta$. With roughly 113 origins per horizon, the accuracy-versus-horizon curve gains real replication.

Scoring rules. Three, because they probe different things. *Log predictive density* rewards sharp, well-located densities and punishes overconfidence severely. *CRPS* is computed with the exact ensemble identity $\text{CRPS} = \frac{1}{N} \sum_i |x_i - y| - \frac{1}{N^2} \sum_{i < j} |x_i - x_j|$, evaluated in $O(N \log N)$ by sorting; it is in kWh and so is interpretable. *Empirical coverage* of central 50% and 95% intervals, plus PIT histograms, tests calibration directly.

One refinement is worth noting. The h -step predictive density is not available in closed form, but it need not be estimated by kernel smoothing: conditional on a posterior draw and the simulated path up to $h - 1$, the final innovation is exactly t_ν . Marginalising only that last step gives the Rao-Blackwellised mixture

$$p(y_{T+h} | \mathbf{y}) \approx \frac{1}{S} \sum_{s=1}^S t_{\nu^{(s)}} \left(z_{T+h}; \mu_{T+h}^{(s)} + \sum_{\ell} \phi_{\ell}^{(s)} \varepsilon_{T+h-\ell}^{(s)}, \sigma^{(s)} \right), \quad (6)$$

which is far more accurate than smoothing a predictive sample and costs nothing extra.

6 Results

6.1 Convergence and the fitted family

Seven specifications were fitted to the full series, all sharing priors, data, and likelihood index. Four chains of 40,000 iterations, 10,000 discarded, thinned by 5, giving 24,000 draws. Diagnostics passed throughout: worst $\hat{R}$ across all seven models was 1.011 and the minimum bulk ESS was 399 (both in M2, for ν , the hardest parameter); the headline model M7 had worst $\hat{R} = 1.007$ and minimum ESS 1,072.

ID	Specification	elpd _{WAIC}	p _{WAIC}	Δ	Purpose
M5	$J=2$, AR(2)+ t , log	−4594.0	18.1	0.0	longer error memory?
M4	$J=3$, AR(1)+ t , log	−4613.2	19.4	−19.2	third harmonic too far?
M3	$J=2$, AR(1)+ t , log	−4622.3	17.2	−28.4	adds harmonic 2
M2	$J=1$, AR(1)+ t , log	−4636.2	15.3	−42.3	mandated starting point
M7	$J=2$, AR(1)+ t , kWh	−4656.8	16.0	−62.9	isolates the log transform
M6	$J=2$, AR(1)+Gauss, log	−4657.6	16.0	−63.7	isolates heavy tails
M1	$J=1$, iid Gaussian, log	−5006.8	12.6	−412.9	naive baseline

6.2 WAIC and the holdout disagree, and the holdout wins

Each modelling decision was isolated as a paired comparison under both criteria. Design B differences are per-day mean log predictive densities with paired standard errors.

Decision	WAIC Δ elpd	verdict	Holdout Δ lpd/day	verdict
AR(1)+ t vs iid Gaussian	+370.6 $\pm$ 36.3	supported	+0.0139 $\pm$ 0.0016	supported
Second harmonic ($J=2$ vs 1)	+13.9 $\pm$ 5.7	supported	+0.0366 $\pm$ 0.0035	supported
Third harmonic ($J=3$ vs 2)	+9.1 $\pm$ 4.5	supported	+0.0457 $\pm$ 0.0017	supported
t vs Gaussian, given AR(1)	+35.3 $\pm$ 11.9	supported	+0.0094 $\pm$ 0.0011	supported
AR(2) vs AR(1)	+28.4 $\pm$ 10.6	supported	−0.0232 $\pm$ 0.0011	rejected
Log scale vs kWh scale	+34.5 $\pm$ 17.7	inconclusive	−0.0772 $\pm$ 0.0040	kWh wins

The two rows in bold are the substantive surprises. WAIC favours AR(2) by nearly three standard errors, yet AR(2) forecasts *worse* out of sample by more than twenty. The explanation is the one the exploratory analysis anticipated: the extra lag is fitting the absence blocks, which improves one-step-ahead in-sample density but does not generalise. Likewise the log transform, which WAIC cannot decide about, is clearly harmful for forecasting. Accordingly the headline model is **M7**: $J=2$, AR(1) with t innovations, on the kWh scale.

6.3 Answers to the six questions

All figures below are posterior means with 95% credible intervals from M7.

Q1 — variation over the study period. Mean consumption is 25.96 kWh/day [25.41, 26.50] at the record midpoint. The annual cycle is by far the largest source of variation, with a full seasonal range of 18.25 kWh/day [16.48, 20.06]; the weekly cycle contributes about 4.56 kWh; the one-step innovation standard deviation is 5.59 kWh.

Q2 — long-term trend: no evidence. The slope is -0.186 kWh/year $[-0.667, +0.292]$, equivalently -0.72% /year $[-2.57, +1.13]$. The interval straddles zero and $\Pr(\text{slope} < 0 \mid \text{data}) = 0.775$ — suggestive of a mild decline, far from conclusive. Over the whole record the implied change is -0.74 kWh/day $[-2.63, +1.15]$, i.e. under 3% of the mean level. The honest answer is that four years is too short to separate a small trend from the annual cycle, particularly since the record is 3 years 11 months rather than a whole number of years.

Q3 — annual seasonality, and yes the second harmonic matters. The first harmonic has amplitude 8.08 kWh [7.31, 8.85], a peak-to-trough swing of 16.15 kWh, and peaks on **14 January** [9–20 Jan]. The second harmonic has amplitude 2.43 kWh [1.68, 3.19], giving an amplitude ratio $A_2/A_1 = 0.302$ [0.204, 0.405] with $\Pr(A_2 > 0.15 A_1 \mid \text{data}) = 0.999$. Three independent lines of evidence therefore agree that harmonic 2 is important: the amplitude ratio, the WAIC difference ($+13.9 \pm 5.7$), and the held-out log score ($+0.0366 \pm 0.0035$). Substantively, the second harmonic makes the annual curve *asymmetric* — it deepens and delays the summer trough while sharpening the winter peak, which is what a heating-dominated load with a long flat summer floor actually looks like. The third harmonic helps too, by an even larger held-out margin, which is the strongest hint that two harmonics still under-fit the seasonal shape.

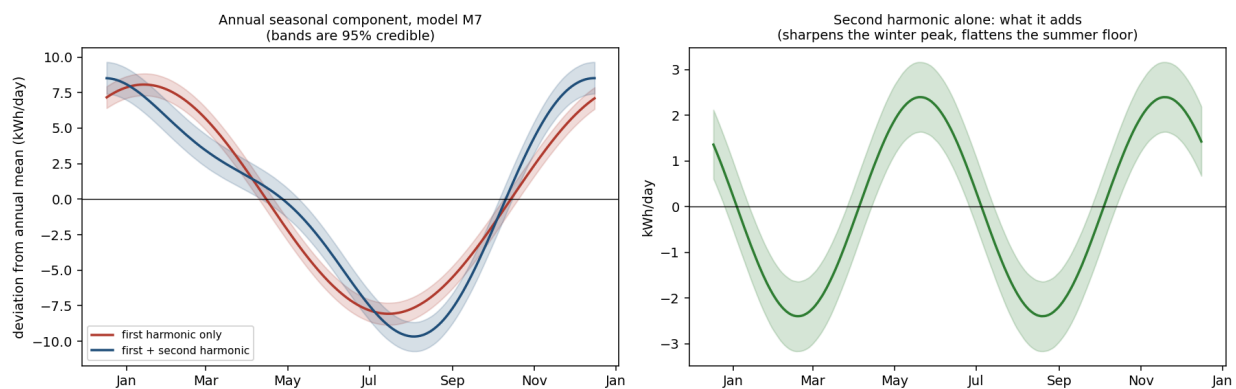


Figure 3: Left: the annual seasonal component with one harmonic (red) and two (blue). Adding the second harmonic deepens the summer trough by roughly 1.5 kWh/day and shifts it later. Right: the second harmonic in isolation.

Q4 — weekly pattern: strong and unambiguous. Deviations from the weekly mean, in kWh/day: Saturday +3.50 [2.67, 4.33], Sunday +3.02 [2.21, 3.85], Wednesday -0.10 , Tuesday -0.41 , Friday -1.12 , Monday -2.15 , Thursday -2.75 $[-3.51, -1.99]$. The weekend-minus-weekday contrast is 4.56 kWh/day [3.71, 5.42] with $\Pr(\text{weekend} > \text{weekday} \mid \text{data}) = 1.0000$ — the clearest effect in the entire analysis, and physically obvious: the house is occupied.

Q5 — predictive accuracy saturates within four days. This is the cleanest result. The width of the 95% predictive interval rises from 28.9 kWh at one day ahead to 31.4 kWh by day seven

and then stops, sitting at 31.2kWh even sixty days out — a total increase of 8%. The empirical curve matches the $AR(p)$ theoretical prediction $\sqrt{\sum_{j<h} \psi_j^2}$ almost exactly, and its ceiling is the stationary ratio $1/\sqrt{1-\phi^2} = 1.079$ [1.057, 1.105] implied by $\phi = 0.375$ [0.323, 0.426]. Calibration holds up: 95% intervals cover 94.5% at $h = 1$ and 95.2% at $h = 7$.

The interpretable conclusion is blunt. Knowing yesterday's consumption buys about 8% narrower intervals, and only for three or four days. Beyond roughly a week, a forecast of this household's demand is seasonal-plus-weekly climatology and nothing more.

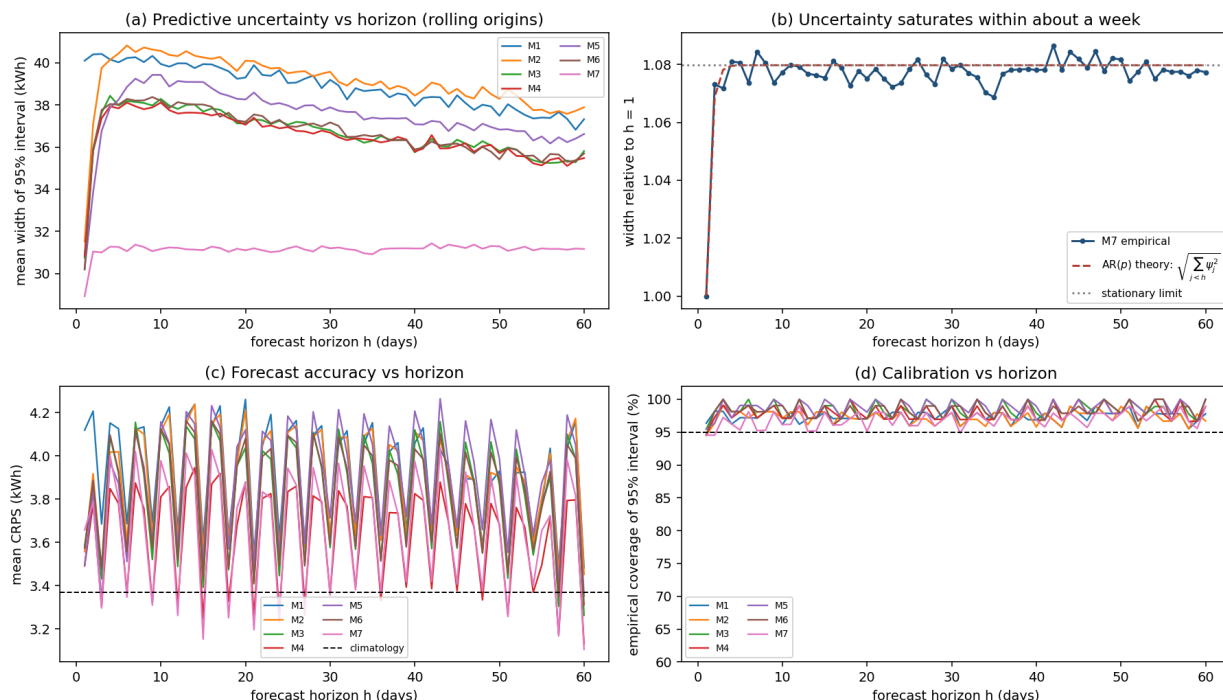


Figure 4: Predictive uncertainty, accuracy, and calibration against forecast horizon. Panel (b) is the answer to Q5: uncertainty saturates by day four and tracks AR(1) theory exactly. The sawtooth in panel (c) is an artefact of origins spaced three days apart, discussed in Section 7.

Q6 — unusual periods. Two diagnostics are needed, for a reason that is itself a finding. One-step-ahead posterior predictive p -values $p_t = \Pr(y_t^{\text{rep}} \leq y_t \mid \mathbf{y})$ flag 31 of 1,408 days at the 1% level, mostly *high* days (23 against 8 low, where 14 of each is expected), concentrated in the first winter: 23 Dec 2006 (79.6 against 38.2 kWh fitted, $z = +7.1$), 3 Feb 2007 (67.2 vs 35.3), 18 Feb 2007 (63.8 vs 33.6). That asymmetry says the model systematically under-predicts extreme winter peaks.

But a one-step-ahead check *cannot* detect a sustained anomaly: once the AR(1) term absorbs a drop, the model expects the low level to continue and subsequent innovations look normal. Since Q6 asks about *periods*, the test statistic must be a block statistic — here the mean of a rolling 14-day window, referred to its posterior predictive distribution under the stationary error process. Of 1,416 windows, 63 fall below $p = 0.01$ and 19 rise above $p = 0.99$. Those windows overlap heavily, so the effective number of independent windows is nearer $1416/14 \approx 101$; even against that denominator, 82 flagged windows is far more than the 2 expected under a correctly specified model. Sustained departures are thus a systematic feature, not sampling noise. Merging overlapping same-direction runs yields thirteen distinct episodes.

From	To	Days	Observed	Expected	Ratio	p
17 Dec 2006	6 Jan 2007	21	42.16	34.57	1.22	0.001
2 Feb 2007	25 Feb 2007	24	36.84	31.14	1.18	0.001
21 Feb 2007	11 Mar 2007	19	25.10	29.79	0.84	<0.001
19 Mar 2007	7 Apr 2007	20	33.54	27.94	1.20	0.003
3 Apr 2007	28 Apr 2007	26	19.99	27.01	0.74	<0.001
21 Oct 2007	11 Nov 2007	22	25.12	30.50	0.82	0.001
19 Dec 2007	2 Jan 2008	15	40.89	34.44	1.19	0.008
17 Feb 2008	9 Mar 2008	22	24.95	29.78	0.84	0.001
1 Aug 2008	5 Sep 2008	36	8.55	17.21	0.50	<0.001
21 Oct 2008	9 Nov 2008	20	25.31	30.29	0.84	0.003
27 Jan 2010	10 Feb 2010	15	37.85	31.15	1.21	0.007
22 Feb 2010	8 Mar 2010	15	23.19	28.94	0.80	0.007
1 Aug 2010	15 Aug 2010	15	9.64	16.30	0.59	0.009

Observed and *Expected* are mean kWh/day over the episode, the latter from the fitted systematic component; p is the smaller tail probability of the most extreme 14-day window in the episode. The thirteen episodes are generated directly into this table by `05_derived.py`, so no figures here are transcribed by hand.

The August episodes are the signature of French summer holidays: in August 2008 the house ran at *half* the consumption the seasonal model expects for five straight weeks, which is a base load with nobody home. The winter episodes run the other way and are almost certainly cold snaps. Both kinds are behavioural or meteorological, and neither is predictable from a calendar — which is precisely the argument for the t likelihood, and precisely the limitation named in Section 8.

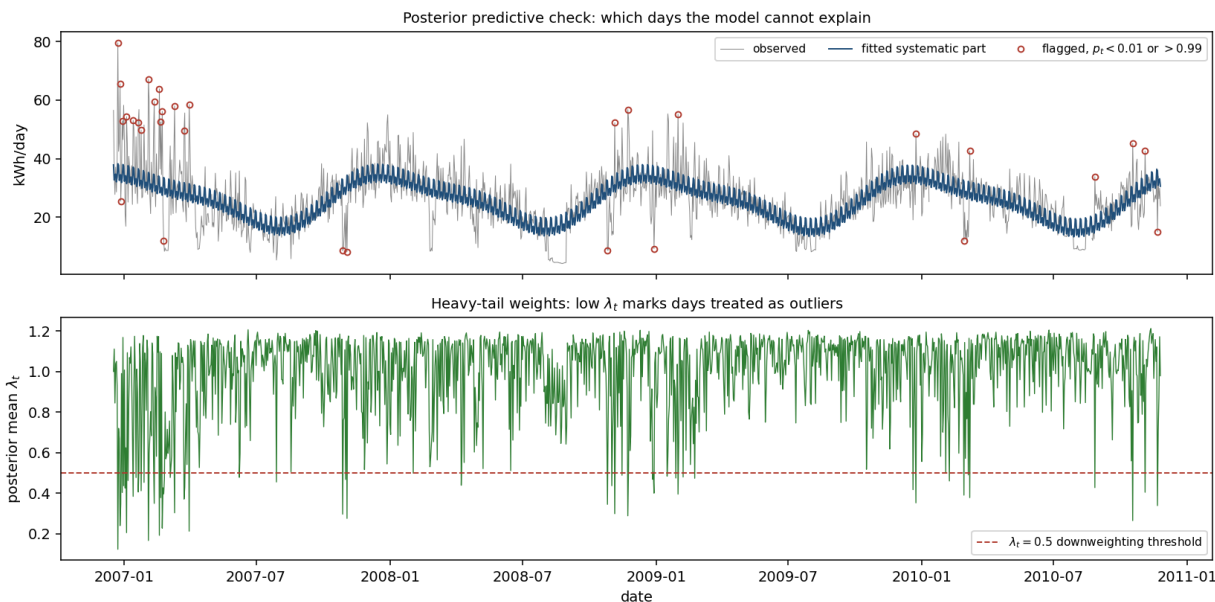


Figure 5: Top: fitted systematic component against observed consumption, with one-step-ahead flagged days circled. Bottom: posterior mean heavy-tail weights λ_t ; low values mark days the likelihood down-weighted.

7 Roadblocks

7.1 Data engineering

1. **Silent parse corruption.** The `?` missing-value code and the day-first dates each fail quietly rather than loudly. Both were handled explicitly, and the sub-meter-versus-total inequality was used as an independent check that parsing was correct (0 violations).
2. **Power versus energy.** `Global_active_power` is kW, not kWh. The division by 60 in (1) is easy to omit and produces a series wrong by a factor of 24 while still looking entirely plausible.
3. **Contiguous missingness creating phantom anomalies.** Because gaps arrive in runs, a naive daily sum turns partially observed days into dramatic low-consumption events, which Q6 would then faithfully report as findings. The 95% coverage rule removes 23 days; without it, several would have contaminated the anomaly table.

7.2 Modelling decisions that came out backwards

4. **The log transform was the wrong default, and I had recommended it.** The initial plan specified a log-scale response on the usual grounds that daily kWh is positive and right-skewed. The residual diagnostics contradicted this (skew $+0.059$ on the kWh scale against -1.450 on the log scale) and the holdout confirmed it decisively (-0.077 nats/day). The cause is specific to this dataset: logging converts the multi-week absence periods from moderate low values into extreme left-tail outliers. Keeping the scale as an explicitly tested decision rather than an assumption is what caught this.
5. **Apparent long error memory was a regime artefact.** The residual ACF at lag 7 is $+0.273$ where AR(1) predicts $+0.023$, and BIC on all days prefers AR(3). Excluding the absence blocks, BIC prefers AR(1) and $\hat{\phi}_1$ drops from 0.58 to 0.38. Fitting the apparent memory with extra AR lags would have been fitting four holidays.
6. **WAIC and held-out forecasting disagreed on AR(2).** WAIC favoured it by $+28.4 \pm 10.6$; the holdout rejected it by -0.0232 ± 0.0011 per day. WAIC on a one-step-ahead conditional factorisation is a weak proxy for forecast skill, and information criteria are known to be unreliable for autocorrelated series. This is why the holdout, not WAIC, is the criterion defended here.

7.3 Implementation and evaluation bugs

7. **Non-comparable WAIC across AR orders.** Models with $p = 0, 1, 2$ naturally evaluate 1,417, 1,408, and 1,399 pointwise terms. Summing different numbers of terms and comparing the totals is meaningless. Fixed by forcing one shared likelihood index set across the whole family. The analogous problem across response scales was fixed with the log-Jacobian.
8. **Misaligned rolling-origin comparison.** Each model initially chose its own valid origins based on its own lag requirements, so paired differences compared forecasts of different calendar days — and crashed on a shape mismatch, which is the good outcome. Fixed by computing one origin set from the strictest requirement in the family and asserting that every model scores identical target days at every horizon.
9. **Dependent horizons in the rolling design.** Origins spaced three days apart mean horizons h and $h + 3$ score nearly the same calendar dates, so adjacent horizons are not independent samples and the accuracy curve oscillates (the sawtooth in Figure 4c). Verified not to be a misalignment bug by checking that the weekday composition is flat across horizons (weekend fraction 0.25 to 0.30). The fix — stepping origins every day and scoring all horizons on one common date set — is identified but not yet applied.

10. **A formatting bug that made CRPS look $100\times$ too large.** Coverage columns were detected with `key.startswith("c")`, which also matches "crps", so CRPS was multiplied by 100 in one table. Harmless to the analysis, but a good illustration of why the same quantity was reported in two places.
11. **One-step-ahead diagnostics cannot answer Q6.** The first anomaly check used sustained runs of low λ_t and found exactly one four-day run, which is wrong — the 36-day August 2008 absence was invisible. The reason is structural: after the AR(1) term adapts, a sustained anomaly produces normal-sized innovations. Q6 required a block-level posterior predictive check with a window-mean test statistic, which then recovered all the expected episodes.
12. **Spurious numerical warnings.** `numpy` on macOS reported divide-by-zero, overflow, and invalid-value warnings from every `matmul`. Confirmed spurious by checking a matrix with condition number 1.52 that raised all four flags while returning exactly correct finite results: Apple's Accelerate BLAS sets IEEE status bits inside its SIMD kernels. Suppressed with a documented filter rather than left to mask real problems.

7.4 Open problem: climatology still wins at long horizons

The one unresolved issue, stated plainly because it qualifies every long-range claim above. A reference forecast built from nothing but day-of-year means (a ± 7 -day window, split by week-end/weekday, from the training years only) *beats every fitted model* on the twelve-month-ahead test:

Forecast	mean lpd	CRPS (kWh)	mean 95% width (kWh)
Day-of-year climatology	− 3.225	3.371	29.89
M7 ($J=2$, AR(1)+ t , kWh)	−3.311	3.666	31.25
M4 ($J=3$, AR(1)+ t , log)	−3.365	3.683	38.08
M3 ($J=2$, AR(1)+ t , log)	−3.413	3.904	38.47
M1 ($J=1$, iid Gaussian, log)	−3.453	3.959	40.56

The climatology is both sharper and more accurate. The working diagnosis is that two harmonics cannot reproduce the true annual shape — in particular the sharp August trough — so the unexplained seasonal structure inflates σ , widening intervals on every ordinary day. Three pieces of evidence support this reading: each added harmonic keeps improving the held-out score (harmonic 3 helps *more* than harmonic 2); the log-scale models are visibly over-dispersed (50% intervals covering 58–68%); and the climatology's advantage comes precisely from being locally flexible in day-of-year while carrying no extrapolated trend.

The planned resolution is to extend the harmonic order to $J = 6$ on the kWh scale, fit a variant without the linear trend (whose extrapolation over a year may itself be damaging), and enter the climatology as a competitor in the rolling design rather than only the single-origin one. Until that is done, the long-horizon forecasts here should be read as a lower bound on achievable accuracy.

8 Limitations

1. **One house, not a market.** A single household's daily load has no averaging across consumers to smooth individual behaviour — one person deciding to do laundry moves the response. Conclusions describe *this* household, and the wide predictive intervals (± 15 kWh around a mean of 26 kWh) are a genuine feature, not a modelling failure.
2. **No temperature covariate.** Heating and hot water drive most of the seasonal signal, and heating demand responds to weather, not to the calendar. The harmonics are a proxy for

average weather on a given day of year, so a cold snap in a mild February is unexplained noise by construction. This is the honest explanation for most of the Q6 anomaly table and the main reason the t likelihood is doing real work. Merging in daily temperature for Paris would be the single highest-value extension.

3. **Trend and season are mildly confounded.** The record spans 3 years 11 months, not a whole number of years, so the seasonal cycle is not perfectly balanced across the trend. This widens the trend interval and is part of why Q2 is inconclusive.
4. **The mandated period is 365, not 365.25 days.** Over a four-year record this drifts by about three days. The brief mandates 365 and that is what is used, but the phase estimates carry this small systematic error.
5. **The absence regime is modelled as outliers, not as structure.** A latent two-state occupancy process (home/away) would explain the blocks rather than merely downweighting them, and would likely fix the long-horizon deficit. It is a substantially larger model and was left as an extension.
6. **WAIC uses a conditional factorisation.** Comparing $\sum_t \log p(y_t \mid \text{past})$ across models with different AR orders compares different factorisations of the joint density. The shared index set makes the arithmetic valid, but the holdout remains the criterion to trust.

9 Reproducing the analysis

Everything runs on `numpy`, `scipy`, `pandas`, and `matplotlib`; no probabilistic programming language is required. Scripts execute in order.

Script	Purpose	Runtime
<code>01_aggregate.py</code>	Parse, aggregate to daily kWh, coverage rule	11 s
<code>01b_transform_check.py</code>	Response-scale and error-structure evidence	13 s
<code>01c_ar_order_check.py</code>	AR order, with and without absence blocks	1 s
<code>02_validate_sampler.py</code>	Recovery of known parameters on simulated data	11 s
<code>03_fit_models.py</code>	Fit M1–M7, convergence, WAIC	6 min
<code>04_holdout.py</code>	Refit on training data, designs A and B, scoring	4 min
<code>05_derived.py</code>	Derived posteriors, anomaly checks, answer figures	5 s

`model.py` holds the sampler, diagnostics, WAIC, forecasting, and scoring rules. Results are written as plain text to `results/` and figures to `figures/`, so every number quoted above is traceable to a file.

Data source. UCI Machine Learning Repository, Individual Household Electric Power Consumption, <https://archive.ics.uci.edu/dataset/235/>.